Do Knowledge- and Reasoning-Oriented Benchmarks Induce Stable Representation Signatures in LLMs?

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Abstract

Internal signatures could support monitoring, adaptive computation, and mechanistic analysis, but benchmark prompts can be separable without any shared computational distinction. We test whether a prespecified knowledge/reasoning label defines a reusable internal signature in five contemporary 7B to 31B models. We treat the ***task*** *as the unit of inference*: probes hold out complete tasks, labels are permuted across tasks, and the best layer and stream are reselected for every permutation. Raw probes recover an eight-versus-eight-subject MMLU partition in all five models (macro-F1 0.78–0.81), but prompt text predicts it nearly as well: under an identical probe configuration, the activation advantage is only 0.017–0.070 macro-F1 and is not significant. At the prespecified 0.01 level, lexical adjustment removes significance while the adjusted states still identify individual subjects, and the selected site changes across task panels and selection rules. Additionally, two released base-training trajectories show no common direction. These pooled layerwise measures recover the task label but identify no transferable knowledge/reasoning signature at the tested sensitivity, which cannot exclude moderate effects. The resulting protocol gives future internal monitors and intervention studies a sequence of tests to pass. Project page: https://parsa-ra.github.io/lmbehave/.

Study at a glance:
Question: Does a knowledge/reasoning benchmark label identify model processing beyond task and prompt identity?
Findings: Raw states predict the label, but their matched gain over prompt text is not significant, and the selected sites do not transfer consistently.
Use: The protocol tests representation signals before they guide monitoring, adaptive computation, or causal intervention.

1 Introduction

Benchmark suites and downstream analyses often use contrasts between knowledge retrieval and multi-step reasoning [Hendrycks et al., 2021, Wang et al., 2024, Sprague et al., 2024, Song et al., 2026]. If these contrasts induced stable internal signatures, they could motivate latent-state monitors [Feng et al., 2025, McKenzie et al., 2025], adaptive-compute policies [Liu et al., 2023], or mechanistic hypotheses [Hendel et al., 2023, Todd et al., 2024]. Decoding a label is not sufficient, however: a model can encode task identity, vocabulary, and format without implementing one shared *knowledge* or *reasoning* computation [Hewitt and Liang, 2019, Belinkov, 2022, Kumar et al., 2022, Sahoo et al., 2026].

These failure modes matter. A monitor driven by topic or format can appear accurate on one benchmark and fail when prompts or tasks change. A layer that maximizes a contrast on one task set can be

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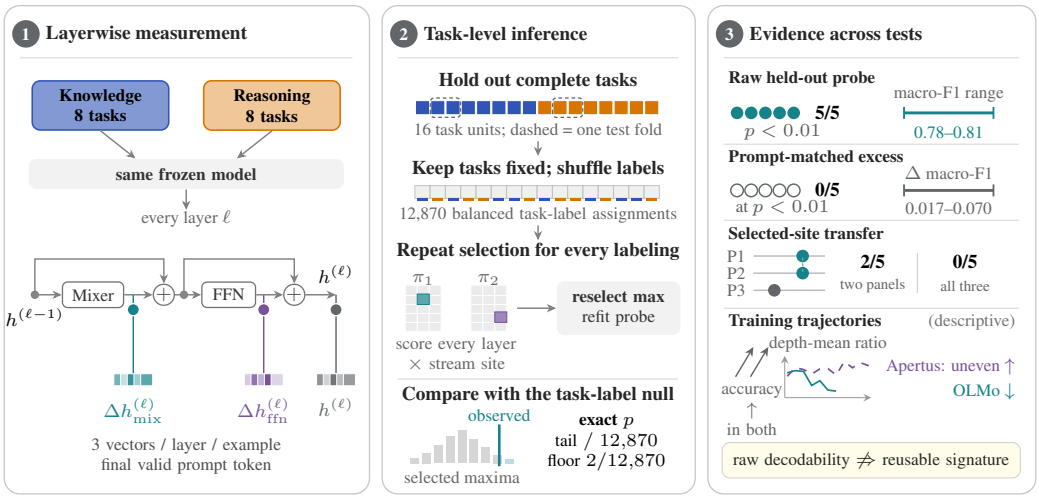


Figure 1: Study design and results. Panel 1: token-mixing and feed-forward updates are read before residual addition, and the residual state after it, all at the final valid prompt token (shading is schematic). Panel 2: held-out-task inference exhaustively permutes task labels (π_1, π_2: two permuted labelings) and reselects the layer–stream site. Panel 3: raw probes recover the partition in every model, but the matched gain over prompt text is not significant, sites do not transfer consistently across the three task panels (P1 MMLU-Pro, P2 MMLU, P3 heterogeneous), and the two training trajectories disagree.

a poor intervention target [Sahoo et al., 2026, Hase et al., 2023]. A candidate signature should therefore generalize across held-out tasks, add information beyond the input, and remain stable across task panels, benchmarks, and site-selection rules (selectors) before it motivates monitoring or causal intervention.

We ask four questions about whether the benchmark label, at this coarse level, can serve as a reusable internal signal. **First**, can intermediate states predict the label of a task that was entirely excluded from probe fitting? **Second**, do they predict it better than the prompt text under a matched estimator? **Third**, does the selected site transfer across panels and selectors? **Fourth**, as a separate training diagnostic, does the measured profile change consistently as accuracy on downstream benchmarks improves? The first three separate recovery of the task label from the stronger evidence that a monitor, router, or intervention target needs. The fourth asks whether the measured profile changes consistently during training, which determines whether it could serve as a training-time signal for future models whose stage schedules differ from those common today.

Our central finding (Figure 1) is a gap between *decodability* and *specificity*. Raw probes recover the label in all five models, but their matched advantage over prompt features is not significant, the selected site depends on the panel and selector, and the two training trajectories do not share a direction. On these task sets, the tested layerwise summaries identify no reusable knowledge/reasoning axis, although moderate effects remain possible. Our contributions are:

- We formulate task-set separation as an exact grouped randomization test. Probes are tested on held-out tasks, labels are permuted across whole tasks, and site selection and probe fitting are repeated for every permutation (Section 2).
- Across five 7B to 31B checkpoints, raw selected-site probes are significant in every model ($p = 0.00155$–0.00420), whereas the matched activation-minus-prompt differences are not ($p = 0.177$–0.440). On an arbitrary control partition, a naive probe that shares tasks between training and test data is significant where the task-held-out probe is not (Section 3.3).
- We test whether the selected site is stable across three task panels and against three covariance-whitened selectors, and we study two released base-training trajectories. Neither the selected sites nor the trajectory directions are stable (Sections 3.1 and 3.3).

Our contribution is an evaluation protocol: task-level holdout and permutation, a matched prompt comparison, transfer tests, and a detection calibration, which together separate task-label decodability from the evidence that reusable internal monitoring requires.

1.1 Related work and scope

Behavioral evaluation, observational decoding, and intervention answer distinct questions: they measure outputs, recoverable information, and output sensitivity under manipulation, respectively [Hewitt and Liang, 2019, Belinkov, 2022, Meng et al., 2022, Conmy et al., 2023]. Latent-state methods can recover propositions or deployment-relevant labels [Burns et al., 2023, Feng et al., 2025, McKenzie et al., 2025], while probes can also learn dataset regularities [Pimentel et al., 2020]. Sahoo et al. [2026] show that format controls weaken a reasoning-mode probe, and they add a causal steering experiment. We test a different binary partition without reproducing their intervention, and we add task-level exchangeability, exhaustive permutation, selection-aware depth inference, five contemporary checkpoints, and two training trajectories. Concurrent contamination-probing work likewise emphasizes nuisance-matched baselines and detection calibration [Braun, 2026].

Task and function vectors make raw task separation plausible [Hendel et al., 2023, Todd et al., 2024]. Our narrower question is whether these task differences align with one transferable knowledge/reasoning direction. Intervention studies report shallower dependence for knowledge retrieval and deeper dependence for reasoning [Song et al., 2026], or selective recall-versus-reasoning effects on controlled puzzles [Fartale et al., 2025]. These task-specific causal findings motivate our test, but they neither imply nor require a benchmark-independent passive signature. Causal tracing restores a clean hidden state in a corrupted run and measures output recovery; activation patching substitutes a component state between paired runs and measures the output change [Meng et al., 2022, Conmy et al., 2023]. Our passive measurements instead generate hypotheses for such interventions.

We report prompt residualization only as a sensitivity analysis, because successful linear removal does not prove concept removal [Kumar et al., 2022]. Effective rank diagnoses variance contraction [Roy and Vetterli, 2007], paired CKA compares geometries [Kornblith et al., 2019], and sparse autoencoders address a different, feature-level question [Huben et al., 2024]. Finally, the benchmark categories are operational proxies: MMLU and MMLU-Pro subjects mix factual recall and inference [Hendrycks et al., 2021, Wang et al., 2024], so our conclusions concern the named task sets and this measurement protocol.

2 Problem Formulation and Method

2.1 Task-level setup and representation sites

Let

$$\mathcal{T} = \mathcal{S}_K \dot{\cup} \mathcal{S}_R, \qquad y_t = \mathbf{1}[t \in \mathcal{S}_K], \qquad \mathcal{D}_t = \{x_{ti}\}_{i=1}^{n_t}, \tag{1}$$

where $\mathcal{S}_K$ and $\mathcal{S}_R$ are the sets of knowledge-oriented and reasoning-oriented tasks, t indexes benchmark tasks, and i indexes examples within a task. The label is therefore defined per task, and examples from the same task are not independent instances of the knowledge/reasoning contrast. For model m, the supported observation sites are

$$\mathcal{J}_m = \{(\ell, s) : 0 \le \ell < L_m,\ s \in \{\mathrm{Mix}, \mathrm{FFN}, \mathrm{Residual}\}\}.$$

At layer ℓ, Mix is the verified token-mixing contribution immediately before residual addition, FFN is the analogous feed-forward contribution, and Residual is the complete decoder-layer output after the residual updates. Mix can be softmax attention or another learned sequence mixer. We treat the three as separately observed streams *without* assuming that they add up, and we record a branch as missing when its semantics cannot be verified.

Let $H_{ti,j} \in \mathbb{R}^{N_{ti} \times d_m}$ be the hidden sequence at site $j = (\ell, s)$ and a_{ti} its prompt mask. Each prompt is represented by its state at the final valid (non-padding) token:

$$r_{ti} = \max\{u : a_{ti,u} = 1\}, \qquad z_{ti,j} = H_{ti,j}[r_{ti}] \in \mathbb{R}^{d_m}. \tag{2}$$

The primary study measures the prompt (prefill) state at this one position, and a separate descriptive analysis averages valid generated-token states within relative response-position bins. The forward hooks only read activations and leave weights and inference settings unchanged. For architecture mappings and invariance checks, see Appendix A.

2.2 Layerwise estimands

For each site j, a label-blind PCA map P_j retains the smallest basis explaining at least 95% of total variance. Let $\tilde{z}_{ti,j} = P_j(z_{ti,j} - \bar{z}_j)$, with means $\mu_{c,j}$ and sample covariance matrices $\Sigma_{c,j}$ for the two task categories $c \in \{0, 1\}$ given by the label y_t (knowledge-oriented tasks have $c = 1$). The prespecified localization statistic is the PCA-space mean-shift ratio

$$R_j(y) = \frac{\|\mu_{1,j} - \mu_{0,j}\|_2^2}{\mathrm{tr}(\Sigma_{1,j}) + \mathrm{tr}(\Sigma_{0,j}) + 10^{-12}}, \qquad j^\star(y) = \arg\max_{j \in \mathcal{J}_m} R_j(y). \tag{3}$$

Unlike the classical Fisher criterion, this ratio does not whiten features by the inverse covariance, so it need not rank sites by linear separability. It is recomputed exactly from per-task sums for every task-label permutation without inverting a covariance; a regularized covariance-whitened variant is reported as a sensitivity analysis (Appendix D). We report its numerator, denominator, PCA dimension, and within-category effective rank so that variance contraction remains visible.

Let the folds $F_1, \ldots, F_K$ partition the complete tasks. For fold k, preprocessing and the classifier are fitted only on examples from tasks outside F_k, and pooling the out-of-fold predictions defines

$$T_{\mathrm{rep}}(y; j) = \mathrm{MacroF1}\left(\{y_t, \widehat{g}_j^{(-F_{k(t)})}(z_{ti,j})\}_{t,i}\right). \tag{4}$$

Probe fitting holds out tasks, but the site $j^\star(y)$ is chosen from all tasks, so the reported macro-F1 is held out for the probe but not for site choice. The exact test accounts for this choice by repeating it for every permutation. The matched prompt comparison reduces the activations and the prompt's character-n-gram TF–IDF features to 64 components each, fitted within each fold, then standardizes both and fits the same logistic classifier with the same folds and regularization. Its paired statistic is

$$\Delta(y) = T_{\mathrm{rep}}^{64}(y; j^\star(y)) - T_{\mathrm{text}}^{64}(y). \tag{5}$$

T_{rep} thus tests recoverability, whether activations predict the label, and Δ tests whether they predict it better than the matched prompt text. As a separate sensitivity analysis, lexical adjustment removes, within each fold, the activation variance linearly predictable from 64 TF–IDF components of the prompt (lexical-64).

2.3 Exact task-label inference and study questions

For $q = |\mathcal{S}_K|$, the conditional randomization space is

$$\Pi_q = \left\{ y' \in \{0, 1\}^{|\mathcal{T}|} : \sum_t y'_t = q \right\}. \tag{6}$$

For statistic S, exhaustive task-label inference uses

$$p_S = \frac{1}{|\Pi_q|} \sum_{y' \in \Pi_q} \mathbf{1}[S(y') \geq S(y)]. \tag{7}$$

For every y', grouped folds are rebuilt, $R_j(y')$ is recomputed, $j^\star(y')$ is reselected, and the probe is refit. With eight tasks per category, $|\Pi_8| = \binom{16}{8} = 12{,}870$. Because the statistics are symmetric in the two class names, the smallest attainable p-value is $2/12{,}870$, well below the prespecified threshold of $p < 0.01$, which was fixed in the frozen analysis manifest before the primary panel's results were summarized; in the five-versus-five panels only the floor $1/126 = 0.00794$ lies below it. Benjamini–Hochberg correction is applied across all layer–stream sites of a model, and the maximum test accounts directly for site selection [Westfall and Young, 1993]. Secondary metrics (RBF probes, cosine silhouette, covariance alignment, paired CKA, and effective rank) are defined in Appendices C–D, and Appendix Figure 4 illustrates what each statistic responds to.

This formulation separates four questions. *Recoverability* asks whether $\max_j R_j(y)$ or $T_{\mathrm{rep}}(y; j^\star(y))$ is significant under Equation 7. *Prompt excess* asks the same of $\Delta(y)$. For panels p, p', we describe localization transfer by

$$A_m^{p,p'} = \mathbf{1}[j^\star_{m,p} = j^\star_{m,p'}], \qquad \rho_{m,s}^{p,p'} = \mathrm{corr}_\ell(R_{m,p,\ell,s}, R_{m,p',\ell,s}). \tag{8}$$

Table 1: Means across three paired example blocks at the first and final checkpoints. Arrows point from the first to the final checkpoint. Figure 2 shows every checkpoint with sample standard deviations.

Model (checkpoints)	$\mathrm{Acc}_K \uparrow$	$\mathrm{Acc}_R \uparrow$	D_{Mix}	D_{FFN}	D_{Residual}
OLMo 3 7B (7)	$0.283 \to 0.585$	$0.279 \to 0.460$	$0.197 \to 0.123$	$0.291 \to 0.145$	$0.275 \to 0.156$
Apertus 1.0 8B (11)	$0.260 \to 0.621$	$0.254 \to 0.433$	$0.174 \to 0.235$	$0.273 \to 0.337$	$0.252 \to 0.323$
Reference	0.250 chance	0.250 chance	0: equal projected means		

Finally, for training checkpoint c, the depth mean

$$D_{m,s}(c) = \frac{1}{L_m} \sum_{\ell=0}^{L_m-1} R_{m,c,\ell,s} \tag{9}$$

tracks whether the measured profile changes consistently with accuracy along a released trajectory.

All four test specified task partitions under a frozen design. Semantic purity, causal specialization, and pruning safety lie outside their scope.

3 Experiments

We begin with the longitudinal analysis, which holds model identity fixed, and then we test specificity and transfer across five contemporary instruction-tuned models.

3.1 Longitudinal training studies

A stable task-set distinction need not strengthen with accuracy, so association with training is informative but not necessary. We ask the narrower empirical question of whether the measured profile changes systematically as performance improves. OLMo 3 and Apertus 1.0 release checkpoints from documented, staged base-training runs [Team OLMo, 2025, Hernández-Cano et al., 2026], which lets us study this within one model at a time. We evaluate seven OLMo 3 7B and eleven Apertus 1.0 8B checkpoints, holding examples, prompts, tokenization, pooling, precision, code, and analysis fixed. Within each trajectory, every checkpoint uses the same three non-overlapping blocks of 32 examples from each of the same ten MMLU subjects (anatomy, clinical knowledge, world history, professional law, and astronomy versus abstract algebra, formal logic, high-school mathematics, college mathematics, and logical fallacies).

Uncertainty. Means and sample standard deviations are taken across the three paired blocks; they measure sensitivity to example selection within a single training run. Greedy prefill measurement is deterministic, so rerunning generation seeds on the same examples would reproduce the same states. The probe uses all 251 non-observed balanced task-label permutations.

For checkpoint c and stream s, we use the depth summary in Equation 9. Because OLMo step counters reset between stages, we report per-step rates of change only within a stage. Apertus checkpoints are indexed by cumulative training tokens. OLMo's three documented stages are pretraining, midtraining, and long-context extension, whereas the Apertus checkpoints span pretraining stages only, so the two curves cover different stage types. Figure 2 shows every sampled checkpoint, and Table 1 compares the endpoints.

These evaluations retained parsed outputs only. Every OLMo response parses as A–D, and the Apertus 1.0 parse rate is 0.997 at its first two checkpoints and 1.000 thereafter, so the parsing failures reported for Apertus 1.5 (Section 3.4) do not drive the accuracy trends below.

OLMo. From the earliest Stage 1 to the final Stage 3 checkpoint, knowledge/reasoning accuracy rises from $0.283/0.279$ to $0.585/0.460$ while D_s falls from $0.197/0.291/0.275$ to $0.123/0.145/0.156$ (token mixing, feed-forward, residual). The mean-shift ratio rises within Stage 2 but moves in mixed directions within Stages 1 and 3. The other measurements disagree with one another: linear-probe macro-F1 rises modestly ($0.811/0.806/0.814$ to $0.837/0.842/0.849$), whereas RBF-probe macro-F1 ($0.798/0.827/0.844$ to $0.792/0.813/0.807$)

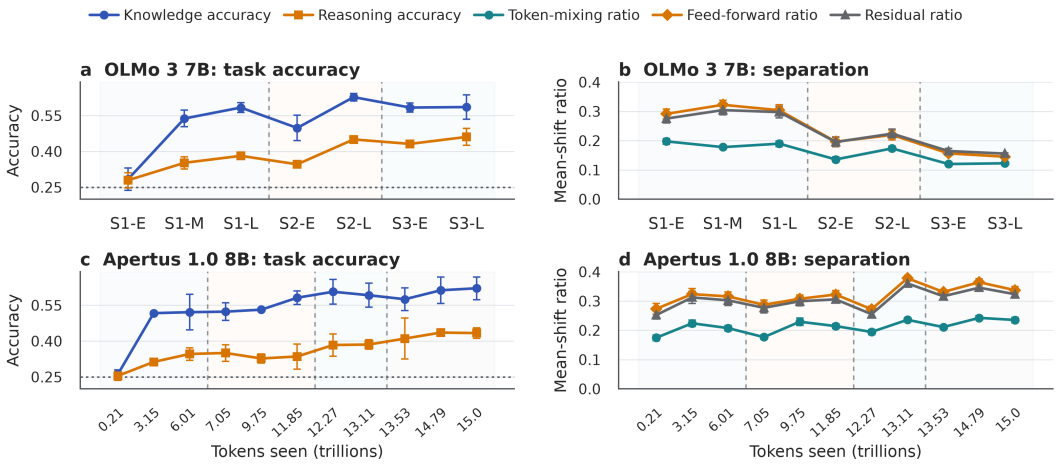


Figure 2: Two released base-training trajectories give different endpoint directions. Accuracy rises in both models. OLMo's depth-averaged mean-shift ratios end below their earliest values, whereas Apertus's end above them. Neither path is monotonic. Error bars are sample standard deviations across three paired example blocks (example-sampling variation only). Dashed lines separate documented training stages. OLMo labels use E/M/L for the early, middle, and late sampled checkpoint within each stage; Apertus labels give trillions of tokens seen.

and silhouette ($0.140/0.194/0.188$ to $0.096/0.108/0.114$) give no sign of increasingly distinct geometries. Within-group effective-rank fractions rise ($0.669/0.654/0.700$ to $0.694/0.683/0.716$), so the depth mean falls without a decrease in the effective-rank fraction.

Apertus. From 210B to 15T tokens, accuracy rises from $0.260/0.254$ to $0.621/0.433$, and D_s rises from $0.174/0.273/0.252$ to $0.235/0.337/0.323$, though not monotonically. At the Stage 3–4 boundary both accuracies rise ($0.581/0.335$ to $0.606/0.383$) while all three mean-shift summaries fall ($0.214/0.322/0.306$ to $0.194/0.272/0.255$). Linear-probe macro-F1 barely moves ($0.798/0.783/0.790$ to $0.806/0.792/0.795$), RBF-probe macro-F1 has mixed endpoint directions, and effective-rank fractions stay near 0.64–0.67, so the depth mean rises without a decrease in this rank summary.

Because the checkpoints are serially dependent and span different training stages, we treat these results as trajectory descriptions rather than independent-sample tests. The two trajectories share no common direction: OLMo depth means fall, Apertus means rise, and both change non-monotonically between checkpoints. Each is one base-training run, and D_s differs from the selected-site statistic, so these observations cannot support population-level conclusions about training trends.

We next test whether contemporary instruction-tuned checkpoints nevertheless show recoverability, incremental prediction beyond prompt text, and transferable localization.

3.2 Cross-sectional design

Models and task panels. We study OLMo 3 7B Instruct [Team OLMo, 2025], Apertus 1.5 8B [Hernández-Cano et al., 2026, Swiss AI Initiative, 2026], Qwen 3.5 9B and 27B [Qwen Team, 2026], and Gemma 4 31B IT [Gemma Team, 2026]. Apertus 1.5, Qwen 3.5, and Gemma 4 are multimodal models evaluated here on text only. Primary comparisons are within model: raw representations and absolute separation values are never pooled across architectures or hidden widths. Full configurations appear in Appendix B.

The heterogeneous cross-benchmark panel places five knowledge-oriented tasks against five reasoning-oriented tasks. All inputs are text, but task family, domain, prompt length, number of choices, and answer type remain confounded with the label, so we use this panel only as a transfer test of the task partition.

We use two distinct single-prompt panels. The first is an MMLU-Pro pilot with five subjects per group and a common ten-option prompt: biology, health, history, law, and economics are con-

trasted with mathematics, physics, computer science, engineering, and philosophy [Wang et al., 2024]. The primary MMLU panel is drawn from a different dataset. It has eight subjects per group in MMLU's four-option format and contrasts anatomy, astronomy, clinical knowledge, world history, professional law, high-school biology, medical genetics, and prehistory with abstract algebra, college mathematics, formal logic, high-school mathematics, logical fallacies, college physics, conceptual physics, and high-school statistics [Hendrycks et al., 2021]. Using 16 subjects increases the number of distinct class-symmetric task-label permutations from 126 to 6,435.

Within each panel, all tasks share one prompt, parser, and generation procedure, but the two panels differ in dataset, subject composition, prompt text, and answer space. The category labels are comparative proxies: clinical and physics questions can require both acquired facts and inference. The primary condition uses 100 examples per task, the largest common count among the selected MMLU subjects, and requests one answer label with a four-token ceiling. We use these protocol-specific scores only to check that the evaluation elicits valid task responses.

We test cross-panel transfer because a reusable monitor or intervention target should work beyond one task panel. A failure to replicate limits that use.

Experimental controls. Across models we fix dataset and model revisions, document IDs, raw task prompts, preprocessing, batching policy, numerical precision, backend, generation settings, and seeds; the batch size is fixed within each model (Table 3). Each instruction model uses the chat template shipped with its tokenizer, so the fully rendered input is model-specific, and Qwen 3.5 and Apertus 1.5 run with their templates' thinking mode disabled. Primary runs use deterministic decoding. Unavailable output lengths are recorded as missing values, and the primary prefill analysis does not condition on downstream correctness or response length.

Analysis plan and inference rules. Within each model, the prespecified localization site is the layer–stream site with maximum PCA-space mean-shift ratio $R_{\ell,s}$. A second maximum excludes residual states and selects between the token-mixing and feed-forward updates. We treat either site as exploratory unless it replicates when site selection is also repeated inside each fold, or in an independent model.

Mean-shift maxima, counts of significant layer–stream sites, and raw probes test recoverability. The matched representation-minus-prompt statistic, separate from the prespecified localization statistic, tests prediction beyond prompt text and is the relevant evidence for that narrower claim, and the heterogeneous panel tests transfer. RBF probes, silhouette, covariance alignment, and post hoc pooling or subgroup analyses are sensitivity analyses or descriptive outcomes, and separately selected maxima are not combined.

3.3 Cross-model results

Each model condition has 100 examples from each of the 16 MMLU subjects, or 800 per group at every layer and stream.

Mean-shift maximum. We recompute every layer–stream mean-shift ratio and reselect the maximum under all 12,870 task-label permutations. The exact test rejects the task-label null at $p < 0.01$ for Apertus 1.5 and both Qwen sizes, but not for OLMo or Gemma (Table 2). An alphabetical eight-versus-eight split, used as a negative control, is not significant in any model ($p = 0.147$–0.269). Many individual layer–stream sites are significant, but these sites are dependent, so their counts do not measure model-level effect size (Appendix A). Every primary maximum is a token-mixing or feed-forward site, so the update-only and overall mean-shift maxima coincide in this panel.

Selected-site probe. When the mean-shift site is reselected and the task-held-out linear probe is refit for every task-label permutation, macro-F1 is 0.78–0.81, and the exact test rejects the task-label null in all five models ($p \leq 0.0042$; Table 2). This includes OLMo and Gemma, whose mean-shift maxima are not significant at $p < 0.01$.

Prompt text alone. TF–IDF probes on the recorded input prompts, with vocabulary, weights, and classifier refit inside every held-out-task fold and balanced task-label permutation, reach macro-F1 0.701 for word 1–2-grams ($p = 0.00870$) and 0.768 for character-within-word 3–5-grams ($p = 0.000777$). The task partition is therefore predictable from the prompt text alone under the same task-level inference.

Matched excess over prompt text. With the matched estimator of Equation 5, which reselects the activation site under every permutation, the prompt probe obtains macro-F1 0.755 ($p = 0.00140$). Representation macro-F1 is 0.773–0.825, but the paired advantage is only 0.017–0.070 and is not significant for any model under the 12,870 balanced task-label permutations ($p = 0.177$–0.440; Table 2). For an alphabetical negative-control split with eight subjects per group, representation and excess p-values are 0.228–0.313 and 0.790–0.861.

Across all task-label permutations, the null distribution of Δ has mean 0.006–0.017, SD 0.057–0.067, and 95th percentile 0.108–0.129, and every observed Δ lies inside this distribution (Appendix Table 7). Replacing activation PCA-64 with fold-local 95%-variance PCA gives deltas of 0.025–0.050 that are likewise not significant ($p = 0.257$–0.466), which argues against the fixed activation width of 64 as the explanation. In an across-model sensitivity analysis with Holm correction over the five models in each analysis condition, all raw probes remain significant at $p < 0.01$, whereas no model in the matched-excess or lexical-64 condition falls below 0.05 after correction.

Unit of inference. On the alphabetical negative-control split, an OLMo probe whose training and test sets both contain examples from every task obtains macro-F1 0.802 ($p = 0.0001$ under 9,999 naive sample-label permutations), whereas task holdout gives 0.574 ($p = 0.228$): treating examples rather than tasks as the unit, in both splitting and permutation, inflates both the score and its significance (Appendix D).

Table 2: Primary MMLU inference and cross-panel localization. Shading marks mean-shift maxima with $p < 0.01$. Matched Δ uses identical 64-feature estimators and reselects the site for every permutation; lexical-64 refits the lexical adjustment within each fold. Sites are zero-indexed layer / stream. The smallest attainable p-value is $2/12{,}870 \approx 1.55 \times 10^{-4}$ for MMLU and 0.00794 for MMLU-Pro. Heterogeneous-panel sites are descriptive, and all sites come from the mean-shift selector.

Model	MMLU site / mean-shift ratio ↑	max p ↓	raw macro-F1↑ / p ↓	matched Δ macro-F1 / p ↓	lexical-64 macro-F1↑ / p ↓
OLMo 3 7B	3 / FFN / 0.221	0.0721	0.781/0.00155	+0.0476/0.2897	0.666/0.01197
Apertus 1.5 8B	0 / FFN / 0.284	0.00435	0.804/0.00186	+0.0695/0.1770	0.613/0.06605
Qwen 3.5 9B	2 / FFN / 0.616	0.00109	0.805/0.00186	+0.0468/0.2909	0.689/0.01476
Qwen 3.5 27B	1 / token mixing / 0.667	0.000777	0.796/0.00420	+0.0172/0.4396	0.655/0.04165
Gemma 4 31B	23 / FFN / 0.499	0.0145	0.805/0.00404	+0.0511/0.2968	0.698/0.02020

Model	MMLU-Pro site / p	MMLU site	Mixed-task site
OLMo 3 7B	3 / FFN / 0.0159	3 / FFN	5 / token mixing
Apertus 1.5 8B	0 / FFN / 0.0238	0 / FFN	8 / token mixing
Qwen 3.5 9B	1 / token mixing / 0.0317	2 / FFN	11 / token mixing
Qwen 3.5 27B	11 / FFN / 0.0317	1 / token mixing	5 / FFN
Gemma 4 31B	12 / token mixing / 0.0238	23 / FFN	14 / FFN

Localization. Only OLMo and Apertus select the same mean-shift site in both single-prompt panels, and no model selects the same site in all three (Table 2). Between MMLU-Pro and the heterogeneous panel, per-stream Pearson correlations of the depth profiles range from -0.047 to 0.922, yet no model retains its maximizing site, so a similar depth profile can coexist with a different maximum. A ridge-regularized, covariance-whitened variant of the mean-shift ratio (three ridge strengths; Appendix D) selects a site other than the mean-shift site in all 15 model–ridge conditions, although every whitened maximum is significant after Holm correction ($p_{\mathrm{Holm}} \leq 0.0094$). Localization therefore depends on the selector. The heterogeneous-panel sites are descriptive because site selection was not repeated within their probe tests.

3.4 Sensitivity analyses and causal scope

Prompt-feature adjustment. Regressing six scalar prompt features, such as token count, out of the activations removes every BH-significant linear-probe site in MMLU-Pro (81, 70, 0, 92, and 88 sites per model before adjustment, none after) but leaves significance in four of the five primary MMLU models. Lexical-64 adjustment lowers primary macro-F1 by 10.7–19.1 points and gives $p = 0.012$–0.066: none is below 0.01, four are below 0.05, and 32 components give the same result. The adjustment is partial and cannot establish mediation: the adjusted states retain 31–54% of between-task variance, 57–76% of total variance, and strong 16-way task prediction under stratified example folds (macro-F1 0.765–0.961, chance 0.0625; Appendix D).

Detection calibration. To estimate power, we inject a known shift between two arbitrary task groups along ten random directions at one fixed OLMo site and rerun the lexically adjusted probe. It detects 0/10 directions at standardized mean differences of 0.5–2 and 5/10 at 4, so moderate signals could go undetected (Appendix Table 9). A separate selection-aware control on the alphabetical partition appends a synthetic site and repeats the complete matched-Δ procedure. The synthetic site is selected for none of the ten directions, even at a standardized difference of 4, and none is detected. The fixed-site control therefore cannot calibrate the full selection-plus-probe procedure. The confounded multimodal analysis is also not significant at $p < 0.01$ (Appendix B.5).

Correctness. We omit correctness conditioning because sparse cells and parse validity of 0.573 (Apertus) and 0.761 (Gemma) would confound correctness with format compliance.

Response position. In an OLMo pilot where the model explains after answering (four tasks per group), the largest prompt-to-response increase over all layer–stream–bin candidates (layer 31, token mixing, bin 5 of 8) raises the ratio from 0.193 to 0.479 ($p = 0.0286$, the smallest value attainable in this design and above the 0.01 threshold), while the depth-averaged ratio falls; the pilot is therefore descriptive (Appendix Figure 3).

4 Discussion and Conclusion

4.1 Implications and applications

Our experiments separate a signal that is *decodable* from one that is specific and stable enough to be *reused*. A raw probe can recover the benchmark label while a matched prompt probe performs nearly as well, and a significant maximum can coexist with unstable localization. Reporting only probe accuracy or the largest layerwise contrast would therefore overstate what the measurement supports.

The immediate use is methodological. For monitoring or adaptive computation, the protocol tests whether an internal readout adds information beyond the input and transfers to new task compositions. For benchmark design, it tests whether a semantic taxonomy reflects more than surface form. For mechanistic analysis, it separates sites worth testing causally from maxima that depend on the panel or selector. Training profiles need replication across checkpoints and runs before they can serve as development signals. This work specifies and tests the evidence that a router, monitor, or pruning policy would require; it validates no such system.

4.2 Limitations

The taxonomy entangles demand, domain, format, and difficulty. Task holdout leaves these confounds in place, and 16 tasks are few independent units for a construct-level claim. The single-run trajectories and the small response pilot cannot estimate population variation. Localization remains conditional on the selector and pooling, and mean pooling, capacity-aware probes, and reliable nonlinear probes remain to be tested. The selection-aware synthetic control does not establish useful power even for the largest tested shift, so a non-rejection cannot be read as equivalence or as the absence of a moderate signal. Finally, the design was not publicly preregistered, and hook invariance does not establish deployment overhead or robustness.

4.3 Conclusion

Raw selected-site probes recover the task partition in all five models, and mean-shift maxima are significant in three. However, the matched prompt advantage is not significant, no model reaches the prespecified $p < 0.01$ level after lexical adjustment, the selected sites vary, and the training trajectories disagree. Moderate effects and processing differences remain possible, but on these task sets pooled layerwise measures alone are insufficient evidence for a reusable computational signature. Before downstream use, a stronger measure should add prediction beyond matched inputs, transfer across panels, show calibrated sensitivity, and support causal tests.

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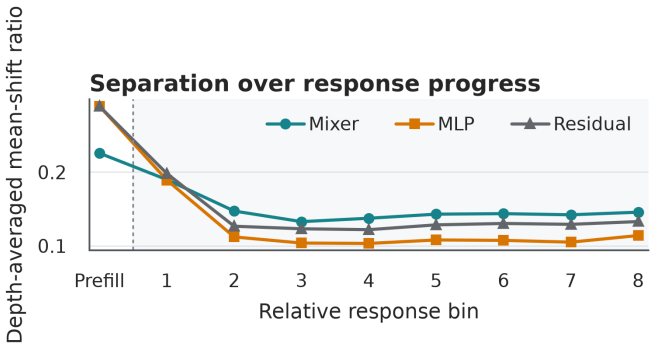


Figure 3: OLMo 3 normalized-response pilot (four tasks per group). The depth-averaged mean-shift ratio falls across response bins, although the reselected local maximum (not plotted) rises $0.193 \rightarrow 0.479$ ($p = 0.0286$, the smallest attainable value). Effective-rank fractions rise. Bins are fractions of the response length rather than semantic reasoning steps.

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A Implementation and Verification Details

The reference implementation is an opt-in component named `RepresentationDynamicsTracker` in our extension of `lmms-eval`. PyTorch forward hooks are attached to one identified decoder stack and return no replacement value. Three hooks per zero-indexed layer capture the token-mixing update, feed-forward update, and residual state. The implementation raises an error when it cannot identify one decoder stack unambiguously, records whether each update is a verified residual contribution, uses the attention mask for padding-aware pooling, and removes all hooks during cleanup.

For OLMo 3 and dense Gemma 4, the verified residual contributions are taken after post-attention and post-feed-forward normalization. Apertus 1.5 and Qwen 3.5 expose verified token-mixing and feed-forward updates directly. The term "token-mixing update" also covers architectures whose sequence mixer is not softmax attention. Gemma 4 applies a learned scalar to the complete layer output after the residual additions, so its adjacent residual change differs from the sum of the two recorded updates. The maximum relative adjacent-state reconstruction error is 0.011–0.042 for OLMo, Apertus, and both Qwen sizes, versus 14.7 for Gemma. This is consistent with the documented Gemma scalar, and no analysis assumes that the three streams are additive.

Response-trajectory validation. The hooks can retain one vector per example, layer, stream, and generated-token update. Absolute positions are distinguished from prefill and from one another. For relative-position analysis, every example's uncapped valid updates are split into eight equal-width bins and averaged within each bin. The analysis reselects the maximum decode-minus-prefill mean-shift change over layer, stream, and relative bin under every task-label permutation. The direct-answer protocol yielded only the first generated-token update: prefill predicts the answer token, while the next recorded state is conditioned on that token. Longer trajectories therefore use a separate, matched explanation protocol with task-specific coverage requirements. The implementation associates updates with each generated sequence and excludes positions at and after EOS or padding. The reported pilot uses uncapped batch-size-one capture, for which no second sequence remains active after EOS. Its first relative bin includes the common answer marker; excluding that marker would require a separately validated token-boundary rule. The pilot contains 178,752 finite vectors from 32 responses, all with verified residual contributions. Raw responses are identical with hooks disabled and enabled, update counts range from 25 to 205 (median 47.5), and every response contributes to all eight bins. These checks establish passive measurement and position coverage for the descriptive pilot, not task-set separation.

For the longitudinal study, the OLMo 3 mapping likewise observes the post-attention and post-feed-forward normalized values added to the residual. The longitudinal source refs and revision prefixes are `stage1-step10000/7717822`, `stage1-step705000/bc2d50b`, `stage1-step1413814/373bad2`, `stage2-step1000/2079c90`, `stage2-step47684/c3c800d`, `stage3-step1000/446ae23`, and `stage3-step11921/4b32e10`.

Apertus 1.5 endpoint and release scope. Apertus 1.5 8B is a suitable second fully open family in principle, and its text decoder exposes verified token-mixing and feed-forward residual contributions. As of 10 August 2026, however, the public release provides only its final revision. The model card states that intermediate checkpoints will follow [Swiss AI Initiative, 2026]. We therefore evaluate the endpoint implementation but do not report a longitudinal comparison. Apertus 1.0 and 1.5 are not substituted as two trajectory points because the release bundles continued pretraining, multimodal training, and post-training changes. The endpoint anchor uses the same ten MMLU subjects as the OLMo trajectory and three disjoint blocks of 32 examples per task. All 92,160 vectors and 288 layer–stream measurements satisfy the count, finiteness, and verified-contribution criteria. Knowledge/reasoning accuracy is $0.358 \pm 0.036/0.306 \pm 0.022$ across blocks; depth-mean ratios are 0.204 ± 0.016, 0.241 ± 0.018, and 0.240 ± 0.018 for the token-mixing, feed-forward, and residual streams. After Benjamini–Hochberg (BH) correction, linear-probe macro-F1 is significant at $q < 0.05$ in all three blocks at 87/96 layer–stream cells. The blocks measure example-sampling sensitivity, not training-seed variation. Greedy answers, scores, and same-process forward logits are identical with hooks disabled and enabled. Thus the endpoint establishes repeated recoverability and passive extraction, but it supplies no training-time direction.

Apertus 1.0 trajectory validation. The separate Apertus 1.0 8B release provides token-indexed intermediate checkpoints across its 15T-token pretraining run [Hernández-Cano et al., 2026]. Its documented 8B curriculum uses Stage 1 through 7T tokens, skips Stage 2, and changes the data mixture at 7T, 12T, and 13.5T tokens before the final cooldown. We evaluated 11 immutable checkpoints spanning the interiors and boundaries of these stages under the same tasks, examples, and measurement protocol as the OLMo trajectory. The 33 checkpoint–block conditions contain 3,168 valid layer–stream measurements with complete counts, finite vectors, verified residual contributions, fixed revisions, paired samples, and complement-symmetric statistics, which aggregate to 1,056 checkpoint-level layer–stream summaries. The results are reported in Section 3.1. The later long-context extension, SFT/QRPO alignment, and Apertus 1.5 endpoint are excluded from the pretraining curve because they change multiple training conditions and require separate analyses. At the 210B, 7.047T, and 15T checkpoints, same-process hook checks cover all 32 decoder layers and 96 verified hook sites. All 384 vectors per checkpoint are finite; forward logits and greedy token IDs are bit-identical with hooks disabled, enabled, and removed, and no hooks remain registered after evaluation.

Each evaluation writes a tensor-bearing `representations.pt`, a vector-free `records.csv`, and `metadata.json`. Offline analysis writes `layer_separability.csv` and `report.json`. These filenames are an implementation contract, not part of the scientific contribution. Manifests record cryptographic hashes, source and package inventories, model and dataset revisions, task/sample identity, prompts, generation settings, batch policy, precision, hardware, selected model stack, verified-contribution flags, and failed or skipped measurements.

The MMLU-Pro and heterogeneous-panel study (100 examples per task) contains 100 model–task conditions and 1,320,000 vectors. The reported MuSR conditions use the corrected two-choice adapter, and the complete cross-benchmark panel is evaluated under all 252 balanced task-label permutations. Greedy outputs match with hooks disabled and enabled on two examples for each evaluated model configuration; same-process logit equality is also established for the longitudinal OLMo endpoint. These invariance results apply only to the configurations tested.

The longitudinal extraction contains seven immutable 7B checkpoints and three disjoint example blocks per checkpoint: 21 conditions and 645,120 vectors. Every condition contains 32 layers, three streams, ten tasks, and 32 examples per task. All 21 conditions satisfy count, finiteness, verified-contribution, revision, source, paired-ID, and disjoint-block criteria. The analysis contains 2,016/2,016 valid layer–stream–block measurements. It uses the same estimator family and all 252 balanced task-label permutations as the MMLU-Pro analysis.

Inference implementation. The reported GPU analysis uses one seeded global task order, filtered under each balanced task-label permutation. Complement symmetry is verified numerically for the mean-shift and RBF statistics and holds by construction for the linear probe (Appendix C). The eight-vs-eight-subject primary panel has 12,870 oriented permutations, including the observed partition and its complement; equivalently, it has 6,435 complement pairs. The five model conditions enumerate all 12,869 non-observed permutations for the primary mean-shift maximum. The five-vs-five-subject MMLU-Pro analysis analogously uses 252 oriented permutations, of which 251 are non-observed. Nonlinear inference uses the same examples, folds, and permutations as its corresponding linear analysis. All reported activation analyses use RAPIDS cuML 26.06 with float64 arithmetic; except for the 64-component matched comparison, probes operate in a 95%-variance PCA space fitted on the training tasks in each fold, and the prompt-only baselines use scikit-learn (Appendix D). Shared-PCA and raw-space scikit-learn activation analyses constitute different estimator families and are not combined with these results. Two processes per model–panel condition run concurrently under NVIDIA MPS.

In the primary panel, the selection-aware mean-shift p-values are 0.0721, 0.00435, 0.00109, 0.000777, and 0.0145 for OLMo, Apertus 1.5, Qwen 9B, Qwen 27B, and Gemma 4. Across layer–stream cells, BH-adjusted significance occurs in 90/96, 95/96, 95/96, 191/192, and 150/180 cells, respectively. The exact p-values for these cell-count statistics are 0.00155, 0.00684, 0.00311, 0.00124, and 0.00824. Because layer–stream cells are dependent, these counts are diagnostics rather than model-level effect sizes.

The selected-site linear-probe analysis performs 6,435 probe refits per model, one for each label-complement pair, while preserving all 12,870 oriented task-label permutation scores. Accuracy and macro-F1 are complement symmetric by construction in all five reports. The macro-F1 randomization p-values are 0.00155, 0.00186, 0.00186, 0.00420, and 0.00404 in the model order above. This is an exact test of the prespecified composite statistic, not a causal or prompt-adjusted result.

Repeated RBF fits do not reproduce all scores exactly, so threshold-adjacent RBF results are treated as numerically unstable rather than as evidence of a qualitative linear–nonlinear contrast. Silhouette scores are computed from explicit batched cosine distances.

B Experimental Details

B.1 Cross-model configurations

Table 3: Evaluated conditions on one 94.5-GiB GH200. All use deterministic prefill extraction, final-valid-token pooling, and 100 examples per task. Batch is the extraction batch size, not the number of analyzed examples.

Model	Revision	Layers	Storage	Batch
OLMo 3 7B Instruct	`6e5971d`	32	float32	32
Apertus 1.5 8B	`a411d83`	32	float32	24
Qwen 3.5 9B	`c202236`	32	float32	20
Qwen 3.5 27B	`fc05dae`	64	float32	8
Gemma 4 31B IT	`842da37`	60	float32	2

B.2 Assets and use terms

We download public model and benchmark releases at the recorded revisions and do not redistribute their weights or benchmark text. The release metadata identify OLMo 3, Apertus 1.0/1.5, Qwen 3.5, and Gemma 4 as Apache-2.0 releases. MMLU and MMLU-Pro are MIT; GSM8K is MIT; ARC is CC-BY-SA-4.0; Natural Questions Open is CC-BY-SA-3.0; MuSR is CC-BY-4.0; HellaSwag is MIT; and SuperGPQA is ODC-BY with additional attribution duties for incorporated sources. The mirrored TriviaQA and MedQA dataset cards do not state a license. We use them only through the public evaluation interface, do not redistribute their content, and do not claim that public availability supplies redistribution rights. The implementation builds on MIT-licensed `lmms-eval`, Apache-2.0 Transformers and RAPIDS, and BSD-licensed PyTorch and scikit-learn. These statements sum-

marize the source metadata; redistribution remains subject to the applicable licenses, notices, and Apertus acceptable-use terms.

B.3 Measured compute

All reported extraction and GPU analyses ran on one NVIDIA GH200 with 94.5 GiB usable memory. Telemetry spans 1.24 hours for the five-model eight-vs-eight-subject extraction, 6.79 hours for its selection-aware probe, and 2.31 hours for the 21-condition OLMo analysis. The Apertus 1.0 analysis used 23.7 logged GPU-hours, including interrupted computation; this is not a clean-run timing estimate. These are single-device wall-clock intervals, not normalized training FLOPs. The matched prompt comparison, lexical diagnostics, and fixed-site calibration share a 3.00-hour telemetry trace that includes unsuccessful solver computation and therefore is not a clean-run timing estimate. Additional exploratory and validation computation is excluded from these totals.

B.4 Text-only cross-model panels

Before model execution, the protocol records the exact 100 document indices and processed-dataset fingerprint for every task; validation requires exact agreement in every model condition. One panel uses a common MMLU-Pro prompt and answer format: biology, health, history, law, and economics are knowledge-oriented proxies; mathematics, physics, computer science, engineering, and philosophy are reasoning-oriented proxies. MMLU-Pro increases reasoning demand relative to MMLU, but its subject categories still mix both demands [Wang et al., 2024].

A second, deliberately heterogeneous stress test pairs closed-book Natural Questions and TriviaQA [Kwiatkowski et al., 2019, Joshi et al., 2017] with MedQA, SuperGPQA, and MMLU-Pro history on the knowledge-oriented side, and GSM8K, MuSR murder mysteries, ARC-Challenge, HellaSwag, and MMLU-Pro mathematics on the reasoning-oriented side [Cobbe et al., 2021, Sprague et al., 2024, Clark et al., 2018, Zellers et al., 2019, M-A-P Team et al., 2025]. This taxonomy follows benchmark objectives only comparatively: GSM8K and MuSR explicitly target multistep reasoning; no-context Natural Questions and TriviaQA most directly target closed-book retrieval; ARC and SuperGPQA require both knowledge and reasoning; HellaSwag is commonsense continuation inference rather than formal reasoning. The panel therefore tests whole-task transfer under broader task sampling, not construct validity. For MuSR, the evaluated adapter renders exactly two choices for all 500 model–example records; string-encoded answer choices are not split into individual characters.

The evaluated current-generation set is OLMo 3 7B Instruct [Team OLMo, 2025], Apertus 1.5 8B [Hernández-Cano et al., 2026, Swiss AI Initiative, 2026], Qwen 3.5 9B/27B [Qwen Team, 2026], and Gemma 4 31B [Gemma Team, 2026]. We exclude model configurations whose managed reasoning traces are incompatible with the fixed short-answer protocol rather than changing generation rules by model.

Prompt token counts differ between the task sets despite the common text modality. In the MMLU-Pro five-vs-five-subject panel, mean knowledge-task counts span 274–308 tokens and mean reasoning-task counts 229–267, depending on tokenizer. A one-feature logistic probe, with the same task-held-out folds and all 252 task-label permutations, does not predict that panel's label significantly (macro-F1 randomization $p = 0.167$–0.270). These values do not describe the primary MMLU panel.

In the MMLU eight-vs-eight-subject panel, the corresponding means are $179.0/135.3$ for OLMo, $172.0/127.1$ for both Qwen sizes, $172.2/129.5$ for Gemma, and $206.6/163.5$ for Apertus. Prompt length is therefore strongly associated with the prespecified partition even though it was not sufficient by itself in the MMLU-Pro panel. The exact TF–IDF control in Section 3.3 uses the logged task prompt before model-specific chat-template rendering. Its vocabulary, inverse-document frequencies, and classifier are refit within every held-out-task fold and balanced task-label permutation. Significance under this null therefore shows that task content already predicts the prespecified partition; it does not quantify all surface information in the model's rendered input or establish that lexical variation causes the representation result.

The primary MMLU evaluation records retain parsed outputs but not raw generated text when the parser fails. Valid A–D outputs occur in 0.992, 0.573, 1.000, 1.000, and 0.761 of examples for

OLMo, Apertus 1.5, Qwen 9B, Qwen 27B, and Gemma, respectively. The primary representations are measured before generation and remain usable; cross-model behavioral accuracy and failure-conditioned analyses are not interpreted. Task-level validity spans 0.93–1.00 for OLMo, 0.20–0.83 for Apertus 1.5, 1.00 for both Qwen sizes, and 0.16–1.00 for Gemma; the released result tables list every model–task condition.

The primary extraction uses the final valid prompt token during prefill and float32 storage. A descriptive decoding pilot is reported above. Mean-valid-token pooling has not yet been evaluated on the complete panel and is discussed as a limitation. Batch sizes are fixed within model after peak-memory calibration. Representations collected under different batch sizes are analyzed separately because calibration showed small but nonzero changes despite identical generated responses.

B.5 Supplementary multimodal benchmark analysis

We conduct a separate cross-benchmark multimodal analysis with Qwen 3.5 9B/27B and Gemma 4 31B. We restrict the comparison to these current-generation multimodal checkpoints and exclude older Gemma 3 checkpoints to match the model-generation rule used in the primary analysis. This analysis is supplementary and does not alter the text-only primary comparison. The prespecified knowledge-oriented group contains OK-VQA, MedQA, MMMU, AI2D, and GPQA [Marino et al., 2019, Jin et al., 2021, Yue et al., 2024, Kembhavi et al., 2016, Rein et al., 2024]; the capability-oriented group contains POPE, MME, TextVQA, OCRBench v2, and DocVQA [Li et al., 2023, Fu et al., 2025a, Singh et al., 2019, Fu et al., 2025b, Mathew et al., 2021]. These operational groups are not matched constructs. All five capability-oriented tasks use images, compared with three of five knowledge-oriented tasks, and the groups also differ in domain, benchmark family, prompt content, and response format.

We retain 100 examples per task and the same final-prompt-token, three-stream extraction. All 30 model–task conditions have complete counts, finite vectors, verified residual contributions, and valid task prompts. Inference holds out complete tasks and enumerates 252 oriented task-label permutations, or 126 label-complement pairs, so the minimum class-symmetric p-value is $1/126 = 0.00794$. The mean-shift maximum is recomputed under every permutation; each selected-site probe then uses that permutation-specific site. Controls remove, within each training-task fold, activation variance linearly predicted from six prompt-surface variables, task-level image presence, three response-format indicators, a 64-component character TF–IDF basis, or both sets. Image presence does not encode pixels, visual-token count, or image content.

Table 4: Supplementary heterogeneous multimodal analysis. Mean-shift cells report the selection-aware maximum, zero-indexed layer and stream (T: token-mixing update; R: residual state), and exact p-value. Probe cells report held-out-task macro-F1 and exact p-value in parentheses. No result is significant under its exact task-label null at $p < 0.01$.

Model	Mean-shift maximum	Raw probe	Surface+modality	Lexical-64	Combined
Qwen 3.5 9B	1.057, L0-T (0.0238)	0.879 (0.0159)	0.210 (0.960)	0.591 (0.349)	0.151 (0.992)
Qwen 3.5 27B	1.199, L5-T (0.0159)	0.858 (0.0317)	0.415 (0.698)	0.550 (0.373)	0.310 (0.952)
Gemma 4 31B	1.113, L36-R (0.0476)	0.899 (0.0159)	0.538 (0.421)	0.735 (0.0635)	0.336 (0.913)

Table 5: Complementary measurements at the observed mean-shift-selected site. These quantities were not given task-randomization tests in this supplementary analysis and are descriptive. Paired CKA is undefined, rather than zero, because the two benchmark groups have no declared example pairing; covariance alignment supplies an order-invariant comparison.

Measurement at selected site	Qwen 9B	Qwen 27B	Gemma 31B	Reference	Role
RBF probe macro-F1	0.837	0.850	0.799	0.50	Nonlinear prediction; descriptive here
Cosine silhouette	0.282	0.256	0.434	0	Higher means greater distance-based separation
Covariance alignment	0.421	0.375	0.080	1	Lower means less similar unpaired geometry
Within-group effective-rank fraction	0.109	0.326	0.418	1	Lower means a more concentrated spectrum
Within-group stable rank	1.202	1.510	2.701	—	Scale-dependent spectral concentration diagnostic
Paired linear CKA	—	—	—	1	Undefined: zero declared cross-group pairs

The raw macro-F1 values exceed the 0.781–0.805 range in the primary MMLU panel, consistent with additional modality and response-format cues that predict the group label; this is not evidence of stronger construct separation. Their exact p-values are 0.0159–0.0317, and prompt-only word and character TF–IDF baselines reach macro-F1 0.804 and 0.802 (both $p = 0.0159$). Adjustment sharply reduces the representation scores (Table 4); Table 5 lists the complementary measurements at the selected sites. This establishes sensitivity to measured task and prompt differences; it neither identifies a modality-independent knowledge/reasoning signature nor proves that the measured variables causally explain the raw score. The result therefore remains separate from the primary text-only evidence.

C Metric Definitions and Inference

Table 6 summarizes the four measurement families and their reference values, and Figure 4 illustrates them on synthetic data.

Table 6: Four complementary measurement families. Reference values aid interpretation; they are not universal quality thresholds. Colors are redundant with the family labels.

Family	Reference value	Scientific role
Mean shift	0: equal projected means; 1: squared mean shift equals summed within-category variance	Within-model mean separation; numerator and denominator remain visible.
Probe	0.50: balanced chance	Prediction for tasks excluded from probe fitting; task-label permutations define the null.
Geometry	Silhouette 0: no distance advantage; alignment 1: identical normalized geometry	Cohesion and similarity; paired CKA is reported only for actual pairs.
Rank	1: maximally even spectrum	Contraction diagnostic, not separation evidence by itself.

PCA-space mean-shift ratio. For each model, zero-indexed layer $\ell \in \{0, \ldots, L-1\}$, and stream s, PCA retains the smallest basis explaining 95% of variance. With projected category means μ_K, μ_R and component variances $\sigma^2_{K,j}, \sigma^2_{R,j}$, the primary descriptive statistic is

$$R_{\ell,s} = \frac{\sum_j (\mu_{K,j} - \mu_{R,j})^2}{\sum_j (\sigma^2_{K,j} + \sigma^2_{R,j}) + 10^{-12}}.$$

We also retain per-component ratios, the numerator, the denominator, retained dimension, and explained variance. A value of zero means that the projected group means coincide; a value of one means that their squared displacement equals the summed within-group variance. The value is not a classification probability and has no model-independent cutoff. Unlike classical Fisher linear discriminant analysis, the denominator is the summed within-category variance $\sum_j (\sigma^2_{K,j} + \sigma^2_{R,j})$ rather than a regularized inverse-covariance quadratic form. It therefore describes total normalized mean shift but need not rank sites by linear separability. The analysis files retain the legacy field `FDR_trace`; significance is calibrated by task-label permutation rather than by the raw magnitude.

Rank diagnostics. For centered activation matrix A with singular values σ_k, define

$$p_k = \frac{\sigma_k}{\sum_j \sigma_j}, \qquad \operatorname{erank}(A) = \exp\left(- \sum_{k:p_k>0} p_k \log p_k \right).$$

We compute effective and stable rank for all examples and after subtracting the corresponding knowledge- or reasoning-category mean. Effective rank is never used as a divisor: dividing by a falling rank would enlarge the adjusted mean-shift value and amplify the pattern it is intended to diagnose.

Probes and finite task-level permutation tests. The linear probe uses fold-local PCA, fold-local standardization, and balanced logistic regression. The nonlinear probe uses the same preprocessing

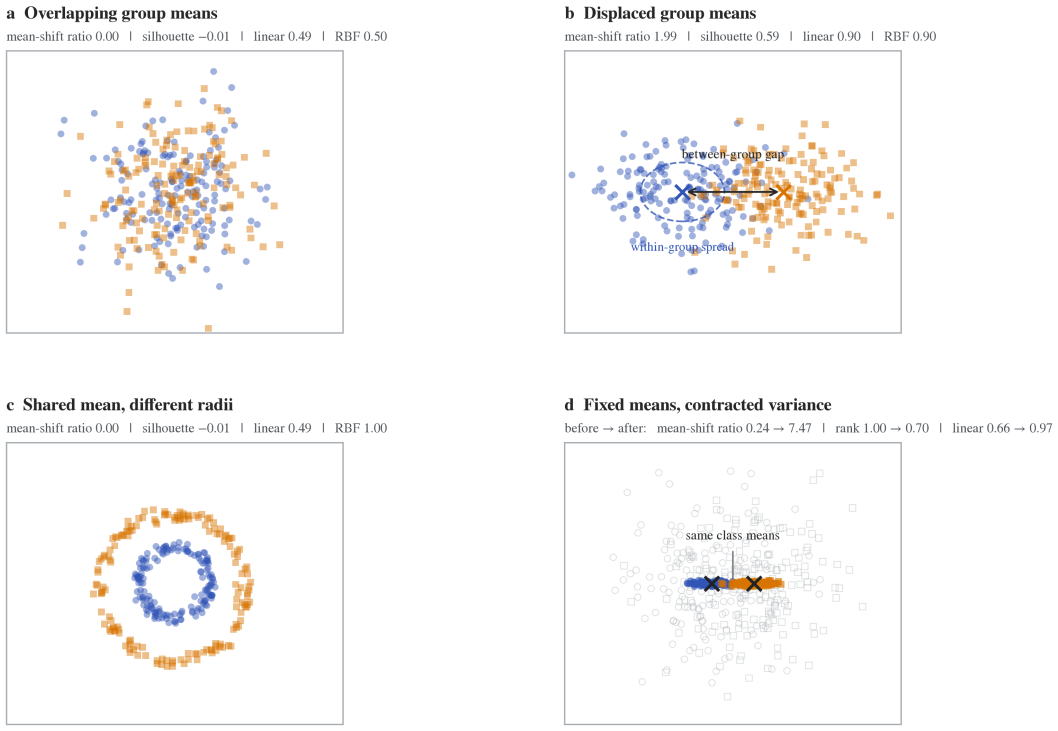


Figure 4: Synthetic metric guide computed directly in two dimensions; no embedding method is used. Blue circles denote K, orange squares denote R, crosses mark class means, and gray hollow points in (d) show the state before contraction. Displaced means raise the mean-shift ratio and both probes (b), whereas equal-mean concentric groups are detected only by the nonlinear probe (c). Holding the means fixed while reducing within-group variance sharply raises the mean-shift ratio (d), illustrating why we report its numerator, denominator, and effective rank separately. Probe values use five complete-group folds with fold-local preprocessing. The examples illustrate what each statistic responds to and define no empirical cutoffs.

and a balanced RBF classifier. Five-fold cross-validation holds out complete tasks from both categories, and pooled out-of-fold predictions yield accuracy and macro-F1. For probe score T, labels are permuted across tasks while preserving the number of tasks per category, and the complete probe is refit:

$$p = \frac{1 + \sum_{b=1}^{B} \mathbf{1}[T_b \geq T_{\text{obs}}]}{B + 1}.$$

Five tasks per category yield 252 oriented balanced task-label permutations, including the observed partition. Accuracy, macro-F1, and the PCA-space mean-shift ratio are invariant to exchanging the two class names, so these permutations form 126 complementary equivalence classes. Enumerating all 251 alternatives therefore gives minimum attainable $p = 2/252 = 1/126$ for these class-symmetric statistics [Ojala and Garriga, 2010]. Mean-shift and RBF label-complement orientations are evaluated independently. For the linear probe, one fitted orientation is reused for its complement because accuracy and macro-F1 are mathematically invariant to exchanging class names; this equality is imposed rather than checked numerically. Mean-shift-selected cell identity is checked independently. A symmetry failure in an independently evaluated statistic makes its exact inference undefined. Benjamini–Hochberg (BH) correction at 0.05 consequently requires at least 16 minimum-p cells when $m = 96$, 29 when $m = 180$, and 31 when $m = 192$ before any cell can be significant after correction. For mean-shift inference, PCA is label-blind and fixed within a layer–stream cell. We aggregate sample counts, activation sums, and squared sums by task, recompute the PCA-space mean-shift ratio for every permutation, and reselect the maximum across all prespecified cells. Each permutation is also treated as observed in turn to obtain its cellwise p-vector and BH-significant-cell count; the observed count is then compared with this finite distribution. The aligned null scores are stored in a compressed analysis file. Benjamini–Hochberg correction is ap-

plied separately to accuracy and macro-F1 across the prespecified layer–stream family [Benjamini and Hochberg, 1995]. Undefined cells retain missing values but still count toward the prespecified family size. Nonlinear probe scores are descriptive unless their own permutation refits are requested.

Geometry. For explicitly paired centered matrices X, Y, linear CKA is

$$\mathrm{CKA}(X, Y) = \frac{\left\|X^\top Y\right\|_F^2}{\left\|X^\top X\right\|_F \left\|Y^\top Y\right\|_F}.$$

It is undefined for unrelated benchmark examples. For unpaired sets we instead report normalized covariance alignment,

$$\rho_{\mathrm{cov}} = \frac{\langle C_K, C_R \rangle_F}{\left\|C_K\right\|_F \left\|C_R\right\|_F}.$$

Cosine silhouette uses $s_i = (b_i - a_i)/\max(a_i, b_i)$, where a_i is mean within-group distance and b_i is distance to the other group [Rousseeuw, 1987]. A zero silhouette means equal average within- and other-group distance; positive values place an example closer to its assigned group on average. We do not import fixed clustering quality cutoffs into this supervised, high-dimensional setting. Silhouette and covariance are computed in the original activation space; CKA is reported only when explicit pairs exist.

D Analysis Hyperparameters and Reporting Fields

The reported linear estimator uses fold-local GPU PCA and standardization followed by class-balanced logistic regression with $C = 1$, ℓ_2 regularization, the cuML quasi-Newton solver, and a 3000-iteration ceiling. The implementation enables cuML's penalty normalization. This is the frozen estimator definition, not an assertion of equivalence to scikit-learn's $C = 1$: the libraries scale their objectives differently and the effective penalty can depend on the training-set size. The nonlinear estimator is a class-balanced GPU RBF SVM with $C = 1$ and scale-adaptive bandwidth. Both probes operate in a 95%-variance PCA space fit only on training tasks in each fold. The descriptive PCA-space mean-shift ratio uses a separate global 95%-variance projection for each layer and stream. The silhouette calculation uses explicit batched cosine distance; the primary eight-vs-eight-subject panel is evaluated exactly, while larger conditions use seeded class-balanced subsampling above the prespecified cap. The primary eight-vs-eight-subject analysis enumerates all 12,870 oriented balanced task-label permutations, comprising the observed partition and 12,869 alternatives. Class-symmetric statistics form 6,435 complementary pairs, including the observed pair; thus 6,434 pairs belong to the null relative to that pair. The MMLU-Pro and heterogeneous-panel linear analyses enumerate all 252 permutations for five tasks per category, and the cross-model nonlinear sensitivity analysis uses the same smaller permutation set; the longitudinal RBF scores remain descriptive because that study did not request nonlinear permutations.

The prompt-only baselines use scikit-learn balanced logistic regression with the liblinear solver, $C = 1$, and ℓ_2 regularization. Word TF–IDF uses lowercased 1–2-grams; character TF–IDF uses lowercased within-word 3–5-grams and at most 100,000 features. The vocabulary, inverse-document frequencies, and classifier are refit inside each task-held-out fold and balanced task-label permutation. These baselines share the inferential unit and permutation set with the representation probe, but not its feature space or GPU estimator.

The prompt-feature sensitivity standardizes its covariates on the training tasks and fits a multi-output Ridge map ($\alpha = 1$) from those covariates to every activation dimension. Training and held-out activations are residualized before fold-local PCA and probing. The map is refit for every fold and task-label permutation. The length-only condition uses the model-tokenizer prompt length; the six-feature condition additionally uses character and line counts, option count, whitespace-token count, and whitespace-token entropy. This operation removes linearly predictable variance, including any construct-correlated variance carried by those covariates; it is a sensitivity analysis, not causal adjustment.

The lexical sensitivity uses the same character TF–IDF definition as the prompt baseline. For every task-label permutation, the unadjusted PCA-space mean-shift profile is used to reselect the layer–stream site. Within each resulting probe fold, the TF–IDF vocabulary and inverse document frequencies are fit on training-task prompts, randomized truncated SVD maps them to 64 components,

and a multi-output Ridge map predicts the activation coordinates. The residual activations then enter fold-local PCA, standardization, and probing. A 32-component run tests dimensionality sensitivity. These logged prompts precede model-specific chat rendering, and a finite low-rank linear map does not exhaust lexical information.

The matched excess test places representations and prompt text under one estimator family. For every balanced task-label permutation, it reselects the unadjusted mean-shift site, maps both feature sets to 64 fold-local components, standardizes them, and fits the same cuML logistic probe with the same grouped folds and regularization. Its statistic is the paired difference in macro-F1, $\Delta = \text{macro-F1}_{\text{representation}} - \text{macro-F1}_{\text{prompt}}$; all 12,870 permutation-specific differences form the exact one-sided reference distribution. Equal feature count and classifier do not make activation and TF–IDF feature spaces semantically equivalent, so a positive Δ remains an incremental prediction result rather than a causal claim.

Table 7: Matched activation-minus-prompt inference and width sensitivity. The null column summarizes 6,434 non-observed label-complement pairs. PCA-95 replaces activation PCA-64 while retaining 64 prompt components; it tests compression rather than capacity matching. Every observed Δ lies within its null distribution.

Model	Matched-64 Δ/exact p	Null mean $\pm$ SD; q95	PCA-95 Δ/exact p
OLMo 3 7B	$0.048/0.290$	$0.016 \pm 0.057; 0.108$	$0.025/0.466$
Apertus 1.5 8B	$0.070/0.177$	$0.011 \pm 0.064; 0.112$	$0.049/0.257$
Qwen 3.5 9B	$0.047/0.291$	$0.011 \pm 0.066; 0.123$	$0.050/0.258$
Qwen 3.5 27B	$0.017/0.440$	$0.006 \pm 0.067; 0.112$	$0.041/0.295$
Gemma 4 31B	$0.051/0.297$	$0.017 \pm 0.066; 0.129$	$0.050/0.330$

Table 7 reports two additional analyses of this comparison. Null summaries use one value from each of the 6,434 null label-complement pairs; the exact test retains all 12,870 oriented task-label permutations. Replacing activation PCA-64 by each model's fold-local 95%-variance activation basis changes neither the sign nor the inference. This unequal-width comparison tests compression sensitivity, not matched capacity.

We also replace the PCA-space mean-shift selector with the regularized whitened mean displacement $\Delta\mu^\top (W + \lambda I)^{-1} \Delta\mu$ in the same label-blind 95%-variance projection. Here W is the permutation-specific pooled within-category covariance and $\lambda = \alpha \ \text{tr}(C)/d$, where C is the label-blind total covariance and d its dimension. For $\alpha \in \{0.01, 0.1, 1\}$, site selection is repeated for every balanced task-label permutation. The vectorized calculation agrees with direct covariance inversion within 2.4×10^{-13} on sampled permutations. The exact test rejects the task-label null for every maximum after Holm correction over the 15-condition grid ($p_{\text{Holm}} \leq 0.0094$), but none selects the PCA-space mean-shift site (Table 8). Thus recoverability is not specific to the PCA-space mean-shift ratio, whereas layer–stream localization depends strongly on the selector and regularization.

The inferential-unit diagnostic uses the alphabetical eight-vs-eight-subject partition and its corresponding layer-0 feed-forward site in OLMo. Random example folds place every task in both train and test and give macro-F1 0.802 ($p = 0.0001$ over 9,999 sample-label permutations). Holding out complete tasks gives 0.574 ($p = 0.228$ over all balanced task-label permutations). The former test is deliberately invalid for a benchmark-category claim; it shows empirically why examples cannot be treated as exchangeable task replicates. It implements the validation contrast, not a reproduction of Sahoo et al.'s datasets, model, or intervention.

Detection calibration uses the observed OLMo selected site and an arbitrary balanced split of the fixed task order, not the knowledge/reasoning labels. Before lexical adjustment, we add a seeded unit-direction shift with standardized mean differences 0, 0.5, 1, 2, and 4. Character TF–IDF fitting, Ridge adjustment, 95%-variance PCA, and the linear probe are then repeated within the grouped folds for every balanced task-label permutation. Ten direction seeds are evaluated at each nonzero shift. This identifies the signal scale detectable by the adjusted fixed-site instrument in one model; it does not calibrate layer selection or measure a model mechanism.

A separate selection-aware calibration uses the alphabetical eight-versus-eight-task partition in OLMo. It appends an injected candidate site to the real-site mean-shift matrix, then reselects the site and refits the matched activation-minus-prompt statistic for every task-label permutation. Across ten seeded directions at each standardized difference 0.5, 1, 2, and 4, the synthetic site is selected for

Table 8: Selector sensitivity on the primary panel. Each entry gives the zero-indexed layer and stream selected by the observed partition; the exact test reselects over all layers and streams for each of 12,870 task-label permutations. These sites are neither causal claims nor a cross-panel transfer test.

Model	Mean-shift site	$\alpha = 0.01$	$\alpha = 0.1$	$\alpha = 1$	Exact p range
OLMo 3 7B	3 / feed-forward	21 / residual	20 / residual	6 / residual	0.00062–0.00078
Apertus 1.5 8B	0 / feed-forward	18 / residual	4 / residual	4 / residual	0.00062–0.00109
Qwen 3.5 9B	2 / feed-forward	30 / residual	30 / residual	30 / residual	0.00093–0.00233
Qwen 3.5 27B	1 / token mixing	25 / token mixing	25 / token mixing	25 / token mixing	0.00062–0.00218
Gemma 4 31B	23 / feed-forward	31 / token mixing	31 / token mixing	22 / token mixing	0.00078–0.00124

Table 9: Fixed-site detection calibration after fold-local lexical adjustment. The injected mean difference is measured in standard deviations of the unadjusted activation projected onto each seeded unit direction. Every condition uses the complete 12,870-permutation null; the zero-shift row is evaluated once because direction is immaterial. Direction seeds are calibration conditions, not model, data, or training replicates.

Standardized mean difference	Directions	Exact $p < 0.01$	Macro-F1 mean [range]
0	1	0	0.472
0.5	10	0	0.475 [0.471, 0.479]
1	10	0	0.483 [0.476, 0.487]
2	10	0	0.513 [0.493, 0.528]
4	10	5	0.724 [0.638, 0.815]

the alphabetical partition 0/10 times and the exact test detects 0/10 directions. This result calibrates the interaction between the unwhitened selector and this particular injected site; it does not imply that no selection-aware procedure can detect a shift of that magnitude.

Grouped folds are rebuilt from the two task classes for every task-label permutation. Across all 12,870 permutations, each of the first three test folds contains two tasks per class and each of the remaining folds one per class; no permuted fold is single-class or less balanced than the observed fold.

For reproducibility, each condition is identified by model and dataset revision, document IDs, task split, prompt, sample count, generation settings, numerical precision, backend, and batching policy. Prompt and output lengths, correctness, semantic stream mappings, skipped records, and analysis failures are retained for sensitivity analyses and auditing. Undefined quantities, including unpaired CKA, are reported as missing rather than set to zero.

Uncertainty sources. We keep four sources distinct. Exact task-label permutations quantify evidence against the probe null. Disjoint example blocks measure sensitivity to selected examples. Repeating fold seeds measures analysis-split sensitivity. Independently trained checkpoints are required for training-seed uncertainty. The current longitudinal table has the first two but only one released training trajectory per model; its mean and standard deviation must not be presented as a confidence interval over model training. Greedy extraction on identical examples is deterministic, so repeating generation seeds would add duplicate observations.

E Use of Large Language Models

General-purpose language-model tools assisted with codebase auditing, implementation review, and editorial revision. The authors remain responsible for the study design, source verification, code, experiments, analysis, and all claims.